

■ 6-MONTH MACHINE LEARNING ROADMAP

Python → Math → ML → Deep Learning → Deployment

by SYNTAX ERROR | syntaxerrorr.com | [@code.abhii07](https://twitter.com/code.abhii07)

6	24	4–5 hrs	1
Months	Weeks	Daily	Project/Week

■ DAILY STUDY PLAN (4–5 hours)

2 hrs	2 hrs	1 hr
■ Learning	■ Coding	■ Dataset Work

■ RULES TO FOLLOW — NO SHORTCUTS

■	Don't just watch tutorials — code along every single video, no exceptions
■	Don't skip math — it's the foundation of ML, understand it deeply
■	Implement algorithms from scratch first — then use Scikit-learn
■	Work with real datasets every week — push everything to GitHub daily

■ TECH STACK YOU'LL USE

■ Python	■ NumPy/Pandas	■ Scikit-learn	■ TensorFlow	■ PyTorch	■ Flask/FastAPI
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■ 6-MONTH JOURNEY

M1	M2	M3	M4	M5	M6
Python + Basics	Math + Stats	Core ML	Advanced ML	Deep Learning	Projects + Deploy

■ MONTH 1 — Python + Basics

Month 1

Python + Basics

Week 1

■ Python Basics — variables, data types, operators ■ Control Flow — if/elif/else, loops (for, while) ■ Functions — defining, parameters, return statements ■ *...and more*

Week 2

■ Data Structures — lists, tuples, dictionaries, sets ■ List comprehensions aur slicing techniques ■ String manipulation aur built-in methods ■ *...and more*

Week 3

■ NumPy — arrays, shapes, operations ■ Broadcasting, indexing, reshaping arrays ■ Mathematical operations on arrays — vectorization ■ *...and more*

Week 4

■ Pandas — Series, DataFrames, basic operations ■ Reading CSV files, exploring data with head/info/describe ■ Data Cleaning — handling missing values, duplicates ■ *...and more*

■ BUILD ■ CSV Data Analysis ■ Simple Data Viz ■ Exploratory Data Analysis

■ MONTH 2 — Math + Statistics (IMPORTANT)

Month 2

Math + Statistics Foundation

Week 1

■ Linear Algebra — vectors, matrices, dot product ■ Matrix operations — transpose, inverse, determinant ■ Eigenvalues, eigenvectors — basic understanding for PCA ■ *...and more*

Week 2

■ Statistics — mean, median, mode, variance, std dev ■ Probability — basic rules, distributions ■ Normal distribution aur z-scores ■ *...and more*

Week 3

■ Correlation aur covariance — measuring relationships ■ Hypothesis Testing — p-values, confidence intervals ■ Data Visualization — histograms, scatter plots, heatmaps ■ *...and more*

Week 4

■ Feature relationships aur dependencies in data ■ Skewness aur kurtosis — understanding data shape ■ Practice week — apply stats concepts to real datasets ■ *...and more*

■ BUILD ■ Statistical Analysis ■ Distribution Analysis ■ Correlation Study

■ MONTH 3 — Core Machine Learning

Month 3	Core Machine Learning Algorithms
Week 1	■ ML Fundamentals — supervised vs unsupervised learning ■ Train-test split, validation, overfitting/underfitting ■ Evaluation Metrics — accuracy, precision, recall, F1 ■ ...and more
Week 2	■ Linear Regression — from scratch aur using Scikit-learn ■ Logistic Regression — classification, decision boundaries ■ KNN (K-Nearest Neighbors) — implementation aur tuning ■ ...and more
Week 3	■ Decision Trees — how they work, pruning ■ Feature importance, tree visualization ■ Ensemble Methods — bagging concept introduction ■ ...and more
Week 4	■ Clustering — K-Means, hierarchical clustering ■ Dimensionality Reduction — PCA basics ■ Project week — combine all algorithms learned ■ ...and more
■ BUILD ■ House Price Prediction ■ Iris Classification ■ Customer Segmentation	

■ MONTH 4 — Advanced ML

Month 4	Advanced ML + Model Optimization
Week 1	■ Feature Engineering — creating new meaningful features ■ Feature Selection — dropping irrelevant features ■ Scaling aur Normalization — StandardScaler, MinMaxScaler ■ ...and more
Week 2	■ Cross-Validation — k-fold, stratified cross-validation ■ Hyperparameter Tuning — GridSearch, RandomSearch ■ Model selection aur comparison strategies ■ ...and more
Week 3	■ Random Forest — bagging, feature importance ■ Gradient Boosting — XGBoost, LightGBM basics ■ Stacking aur ensemble techniques ■ ...and more
Week 4	■ Class Imbalance Handling — SMOTE, class weights ■ Advanced Metrics — ROC-AUC, precision-recall curves ■ Real-world project implementation end-to-end ■ ...and more
■ BUILD ■ Customer Churn Prediction ■ Credit Risk Model ■ Fraud Detection	

■ MONTH 5 — Deep Learning Basics

Month 5

Deep Learning Fundamentals

Week 1

■ Neural Network Basics — neurons, layers, forward pass ■ Activation Functions — ReLU, sigmoid, tanh
■ Backpropagation — how networks learn, gradient descent ■ *...and more*

Week 2

■ TensorFlow aur PyTorch — choosing your framework ■ Building sequential models — Dense layers ■
Loss Functions — MSE, categorical crossentropy ■ *...and more*

Week 3

■ CNNs (Convolutional Neural Networks) — convolution, pooling ■ Image classification tasks with CNN
■ Transfer Learning — using pre-trained models (ResNet, VGG) ■ *...and more*

Week 4

■ NLP Basics — text preprocessing, tokenization ■ RNNs aur LSTMs — sequence modeling ■
Sentiment Analysis task implementation end-to-end ■ *...and more*

■ BUILD ■ Image Classifier (MNIST) ■ Image Recognition App ■ Sentiment Analysis

■ MONTH 6 — Projects + Deployment

Month 6

Portfolio Projects + Deployment

Week 1

■ Model Serialization — saving aur loading models (pickle/joblib) ■ Flask/FastAPI Basics — creating
REST API endpoints ■ REST API for ML models — input → predict → output ■ *...and more*

Week 2

■ Docker — containerizing ML applications ■ Deployment Platforms — Heroku, AWS basics, Render ■
Environment setup aur dependency management ■ *...and more*

Week 3

■ MLOps Concepts — monitoring, versioning, model registry ■ Git aur version control for ML projects ■
Documentation aur code quality — clean readable code ■ *...and more*

Week 4

■ Final Project — end-to-end ML pipeline from scratch ■ GitHub Portfolio setup — showcase all 6
months of work ■ Resume aur LinkedIn — add live project links ■ *...and more*

■ BUILD ■ Web App with ML Model ■ Real-world Dataset Project ■ Capstone Project

■ PROJECT MILESTONE TRACKER

Month	Key Deliverables
Month 1	CSV Analysis, Data Cleaning Projects
Month 2	Statistical Analysis, Distribution Projects
Month 3	House Price Prediction, Iris Classification
Month 4	Customer Churn Prediction, Credit Risk Model
Month 5	Image Classifier (MNIST), Sentiment Analysis
Month 6	Deployed ML Web App, Complete Portfolio

■ RESOURCES — Everything You Need to Become an ML Engineer

■ OFFICIAL DOCUMENTATION

■ Syntax Error	syntaxerrrr.com	ML notes, tutorials, DSA + placement resources — our website
■ Scikit-learn Docs	scikit-learn.org	Complete ML library reference with examples for every algorithm
■ TensorFlow Docs	tensorflow.org	Deep learning framework with official tutorials and guides

■ ONLINE COURSES & TUTORIALS

■ BEST FOR BEGINNERS	Krish Naik — ML Course by Krish Naik youtube.com/@krishnaik06 Best for beginners — clear explanation + code along + real industry insights. Covers Python, ML, DL, NLP, and deployment in depth.
■ DEEP LEARNING	fast.ai — Practical Deep Learning by Jeremy Howard fast.ai Top-down approach — learn by building first, then understand theory. Used by world-class ML practitioners and Kaggle grandmasters.
■ HANDS-ON PRACTICE	Kaggle Learn by Kaggle kaggle.com/learn Free micro-courses on ML, Python, data analysis with hands-on notebooks. Best for practical skill building and real competition experience.

■ DATASETS FOR PRACTICE

■ Kaggle	kaggle.com/datasets	Thousands of datasets and competitions — the #1 place for ML practice
■ UCI ML Repository	archive.ics.uci.edu	Classic datasets for algorithm practice — iris, wine, breast cancer etc.
■ Google Dataset Search	datasetsearch.research.google.com	Find datasets across the web easily — Google-powered discovery

SYNTAX ERROR	Our Website syntaxerrrr.com	Our Instagram @code.abhii07
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"The best ML engineer is not the one who knows the most algorithms, but the one who understands their data best. Start coding today! ■"